

Engineering Reboot 2026

Week 6 – Day 28

Python Virtual Environments (venv)

According to the locked Week 6 workbook, today's theme is **environment isolation**, culminating in a managed Python application running inside its own virtual environment.

Mission

Build the engineering habit of isolating Python projects from the global interpreter so that every project has deterministic dependencies, reproducible execution, and a clean development environment.

Engineering Context

Until now, every Python program you've written has relied solely on the standard library. Real-world software engineering quickly introduces external packages.

Professional developers rarely install packages globally.

Instead they create a dedicated virtual environment for each project.

This prevents:

- dependency conflicts
- version incompatibilities
- accidental system-wide modifications
- unreproducible development environments

Today's objective is not merely learning the commands.

The objective is understanding **why** virtual environments exist and making them part of your normal engineering workflow.

Learning Objectives

By the end of Day 28 you should be able to:

- Explain what a virtual environment is.
 - Explain why environment isolation is important.
 - Create a virtual environment.
 - Activate and deactivate it.
 - Install packages inside it.
 - Verify installed packages.
 - Freeze dependencies.
 - Recreate an environment from requirements.txt.
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Core Concepts

Today's study should cover:

- Python interpreter
 - site-packages
 - pip
 - virtual environments
 - activation scripts
 - requirements.txt
 - dependency management
 - reproducible builds
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Exercises

Exercise 1

Verify Python Installation

Objectives:

- `python --version`
- `pip --version`
- where python (or equivalent)

Validation:

Confirm Python and pip are available.

Exercise 2

Create Virtual Environment

Create a project directory.

Generate a virtual environment using:

```
python -m venv .venv
```

Observe the generated directory structure.

Validation:

The `.venv` directory exists.

Exercise 3

Activate Environment

Activate the environment.

Verify:

```
python --version  
pip --version
```

Confirm the command prompt indicates the active environment.

Validation:

Python executes from the virtual environment rather than the global installation.

Exercise 4

Install a Package

Install a lightweight package such as:

```
colorama
```

Create a short Python script demonstrating its use.

Validation:

Program executes successfully using the installed package.

Exercise 5

Freeze Dependencies

Generate:

```
requirements.txt
```

using:

```
pip freeze
```

Validation:

The installed package appears in the generated file.

Exercise 6

Deactivate Environment

Deactivate the environment.

Verify that:

- prompt returns to normal
 - global interpreter is active again
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Mini Project

Project

Managed Python Application

Develop a small command-line application that:

- executes inside a virtual environment
- imports an external package
- produces useful console output
- includes a `requirements.txt`
- can be recreated by another developer using only:
 - the source code
 - the requirements file

The emphasis is not application complexity but demonstrating a **reproducible engineering environment**.

Validation Matrix

Requirement	Validation
Virtual environment created	PASS
Environment activated	PASS
External package installed	PASS
Application executes	PASS
requirements.txt generated	PASS
Environment recreated successfully (optional verification)	PASS

Evidence Checklist

Capture evidence for:

- Python version
 - pip version
 - Virtual environment creation
 - Activated environment prompt
 - Package installation
 - Program execution
 - requirements.txt
 - Git operations
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Engineering Review

At the end of today's work, consider:

- Why are virtual environments essential for professional software engineering?
 - What risks arise from global package installation?
 - How does `requirements.txt` improve collaboration?
 - How does environment isolation support reproducible builds?
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Notes.md Guidance

Document:

- new commands learned
 - observations about venv structure
 - differences between global and isolated environments
 - package management workflow
 - engineering insights
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Journal.md Guidance

Reflect on:

- what surprised you most about virtual environments
 - how this changes your view of Python development
 - where you anticipate using venv in future projects
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Completion Checklist

- Study completed
- Exercises completed
- Mini-project implemented
- Validation complete
- Evidence captured
- Notes.md updated
- Journal.md updated
- Git committed
- Git pushed
- Git fetch verified

